

AKSHANSH YADAV

Full-Stack & Web3 Developer | Independent Builder | Open to Internships

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SKILLS

Languages: TypeScript, JavaScript (ES2024), Python 3.12, C, SQL, Solidity, Rust (learning), Bash
Frameworks: Next.js 14 (App Router), React 18, Node.js 22, Tailwind CSS, Express.js
Auth & Data: NextAuth.js v4, OAuth 2.0, JWT, Zod, Upstash Redis, REST API design, WebSockets
UI & Viz: Recharts, Lucide React, Canvas API, shadcn/ui, CoinGecko API
CV & Embedded: OpenCV, MediaPipe FaceMesh, Vosk STT, NumPy, pandas, Arduino/ESP32, Embedded C
DevOps: Git, GitHub, Railway, GitHub Pages, GitHub Actions (learning), Docker (learning)
Web3: Solana, Ethereum, Solidity, ethers.js, Anchor framework, DeFi / AMM mechanics

PROJECTS

SwapX — CEX + DEX Crypto Exchange 2026 | [Live ↗](#) | [github/swapx-dex ↗](https://github.com/swapx-dex)
Full-stack trading platform: CEX order-book (market + limit orders, 0.1% fee) and DEX AMM swaps (constant-product formula, 0.3% fee + slippage) for 30+ tokens. CoinGecko prices cached in Upstash Redis (60s TTL); Google OAuth via NextAuth; per-user portfolio + trade history in Redis Hashes/Lists. Deployed on Railway.

Next.js 14 · TypeScript · Upstash Redis · NextAuth.js · Zod · Recharts · Tailwind CSS · CoinGecko API · Railway

Hand Gesture & Voice Controlled RC Car 2025 | [github/gesture-robot ↗](https://github.com/gesture-robot)
Built solo in 4 weeks. Dual-mode control: MediaPipe hand-landmark gestures via webcam + Vosk offline speech-to-text (English + Hindi), both relayed over UDP sockets to ESP32. Designed full circuit from scratch: L298N motor driver, HC-SR04 ultrasonic sensor, LiPo power rail.
ESP32 · Arduino (Embedded C) · OpenCV · MediaPipe · Vosk · NumPy · UDP

FaceMetrics — AI Facial Geometry Analysis 2026 | [Live ↗](#) | [github/face-rater ↗](https://github.com/face-rater)
Evolved through 16 iterations (Python CLI → Tkinter GUI → browser app). Scores 11 facial metrics (jaw width, cheekbone ratio, symmetry, eye area, chin projection, etc.) via 468 MediaPipe FaceMesh landmarks — 100% in-browser, zero server. Webcam capture + auto-brightness enhancement; gender-specific dimorphism weighting.
JavaScript · MediaPipe FaceMesh · Canvas API · Python · OpenCV · NumPy

Space Shooter 2025 Play ↗ GitHub ↗	Interactive Portfolio (Windows OS Style) 2025 Live ↗
2D arcade shooter rewritten from Python/pygame → pure JS Canvas; loads instantly in any browser. Thrust+inertia physics, two AI enemy types, power-up system, particle FX, touch support. JavaScript · Canvas API · OOP · Python/pygame	Draggable-window desktop resume — music player, GitHub heatmap, rotating wallpapers. Pure HTML/CSS/JS, zero frameworks, on GitHub Pages. HTML · CSS · JavaScript · GitHub Pages

EXPERIENCE

Founding Member & Social Media Lead — Simbha Kunj Foundation (NGO) Feb 2021 - Present · Kanpur
Co-founded animal welfare + student support NGO. Led social media strategy and community outreach growth.

Social & Outreach Head — YouTube Channel Oct 2025 - Present	Independent — E-Commerce & Dropshipping 2024 - 2025
Grew audience via content positioning and engagement strategy.	Product sourcing, digital marketing funnels, data-driven sales analysis.

EDUCATION

B.Tech — HBTU Kanpur Aug 2025 - Apr 2029
Harcourt Butler Technical University
Class XII 2024 · 85% overall
Class X 2022 · 93.6% overall

ACHIEVEMENTS

- Qualified JEE Mains 2024 & 2025; JEE Advanced 2024 & 2025
- SOF Olympiads (Cl. 11) — English: Intl. Rank 240
- RC Plane Development — design, assembly & flight testing
- Co-Founded Simbha Kunj Foundation (NGO)

INTERESTS

Web3 & DeFi | AMM & Order-Book Design | Solana & Ethereum | Smart Contracts | Full-Stack Dev
| REST & WebSocket APIs | DevOps & CI/CD | Computer Vision | Embedded Systems | Open Source
| AI & LLMs